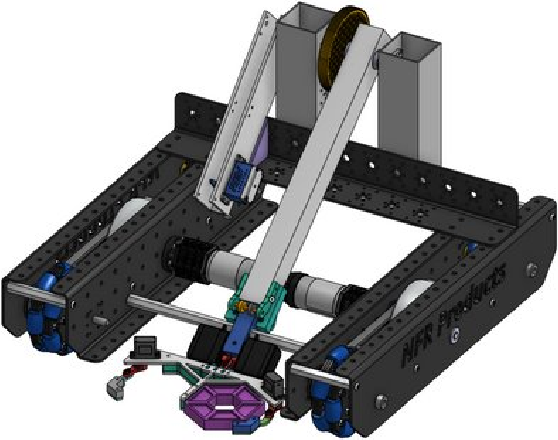
Team #25153
FTC Engineering Portfolio
2023-2024



cartesian

CARTESIAN ROBOTICS

WHO IS CARTESIAN ROBOTICS

We are METU DF Highschool students. As Cartesian Robotics, established in 2020, our main goal is offering a field for self improvemet of our voluntary members.

In season and Off-Season we are showing effort to portray that FRC is “More Than Robots”., meanwhile we are strongly bonded with FIRST’s main mission “Gracious Proffesionalism” We make social responsibility projects for impact.

WHO IS CARTESIAN JR

We are students of METU Development Foundation Private Secondary School. We were founded in 2022 with the mentorship of teachers and experienced students of Cartesian Robotics. The aim of our team is to enable secondary school students to benefit from FIRST values and learn about STEM.

Our team actively participates in STEM competitions suitable for secondary school level, such as Riders Robotics League, FLL and FTC, which we joined this year and became finalist alliance captain.

Team #25153 Cartesian Robotics

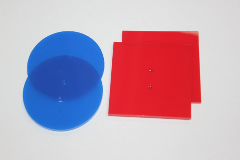
Spreading FIRST!

As Cartesian Robotics, we carried out different activities to disseminate FIRST principles in the first year of FTC in our country.

We also exchanged ideas at meetings and gained efficiency from different ideas.

We held meetings with the Dragon Robotics Team, which was affected by the earthquake disaster in our country this year, and taught them how to use the OnShape program in CAD design.

We have prepared an Alliance Marker and Power Button Plate to distribute to all teams participating in the 2024 FTC Istanbul Off-Season.



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We organized an FRC Kick-Off event with our FRC team at our school. We made informative presentations to our school students and the parents of our team members about FRC, FTC and the concept of STEM in general.

As a team, we met online with Selim Karaağaç, the assembly chief of the Hürjet aircraft at TAI; We learned about manufacturing techniques, design principles, and basic engineering.

In addition, BİIkent Electrical and Electronics Engineering students Ege Doğanay and Özgür Soysal visited our workshop and gave us training on the use of CAD and wiring of the robot.

TEAM PLAN

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	Kim neyi yapacak onu paylaşınız	15					
	Discordda toplanmak, tek tek çizim yapmak	16					
		17					
	Drivetrain ile assembly ve sorunların düzeltilmesi	18					
	İmalat için parçaları hazırlamak	19					
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		27	TÜRK DİLİ ve EDEBİYATI T. MATEMATİK	3.DERS 4.DERS		FİZİK	3.DERS
		28					
		29	Kit'in Gelişi			Kit'in Gelişi	
Orhun kablolama	Drivetrain montajı ve parçaları birleştirme	30	FİZİK DİN K. ve A.B.	3.DERS 7.DERS		KİMYA	3.DERS
Orhun kablolama	Saha yapımı	31	TARİH	3.DERS		COĞRAFYA DİN K. ve A.B.	3.DERS 4.DERS
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	FTC Yarışma (DAHA BELLİ DEĞİL)	17					
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TEAM PLAN

As the Cartesian FTC team, we worked with our FRC team every Monday for 2 months.

While our FRC team is preparing for the FRC competition, the FTC team is also gaining experience from them and gaining experience in robot drawing, programming and wiring.

We had the chance to observe our FRC team as they prepared for their own competitions and thus set an example for ourselves.

MECHANICAL DESIGN PROCESS

BRAINSTORMING:

The first step in our mechanical design process is brainstorming. For this step, we brought our entire team to a meeting and created an environment where everyone shared the first ideas that came to their mind.

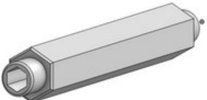
So we came up with a few design ideas that we think might work.

PROTOTYPING:

As the second step of our process, we went through the prototyping phase, which you can see on the pages below. At this stage, we decided on the designs of the subsystems of the final version of our robot by building simple versions of the ideas we created by brainstorming both in our CAD program and in the workshop.

PROBLEM SOLVING:

The final stage of our mechanical design process is problem solving. At this stage, we solved the problems that arose while building the robot with the ideas we found and finalized our robot.



MECHANICAL DESIGN PROCESS

During the problem solving phase, we encountered problems other than robot design. The shaft of one of our robot's main wheel motors broke two days before the competition.

Therefore, we quickly discussed it with our mechanical team and redesigned the shaft sheath by changing the shaft thickness. In this way, we rebuilt our motor shaft with a different size HEX profile in our workshop.

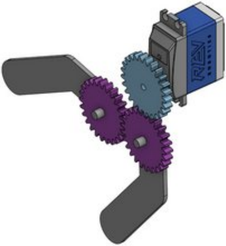
MENTORING

Our team was mentored by a team of experienced members of our FRC team and our teachers.

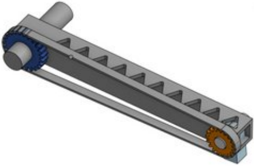
Our mentors helped us solve our problems in different areas during the season, from problem solving stages to determining strategies. They shared their experiences with us

PROTOTYPES/ITERATIONS

First iteration: "PRIMAL GRAB"



Our first idea was to build a simple gear-based grabber for the transformation of the gamepieces.



We intended on using a simple pulley-belt design for the rotation of our grabber.

But we found out that the pulley mechanism didnt work after constructing a concept-demo of it.

So instead we got rid of our pulley and decided on using just a gear mechanism for the rotational movement.

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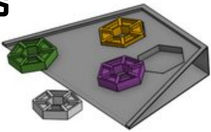
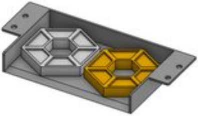
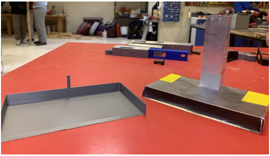
PROTOTYPES/ITERATIONS

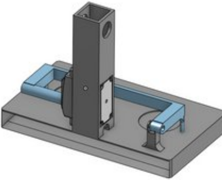
SECOND ITERATION: “SWEEPIN’ ”



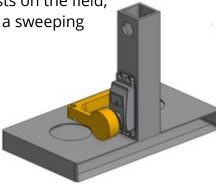


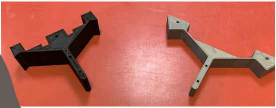
We had multiple designs based on intaking the gamepieces by a sweeping motion.





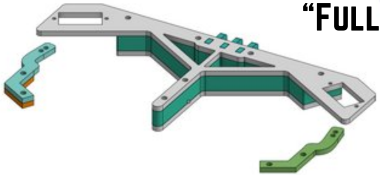
But after carefull consideration and multiple tests on the field; we concluded that using a grabber, rather than a sweeping mechanism was more applicable.





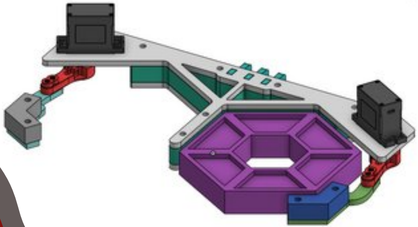
PROTOTYPES/ITERATIONS

THIRD ITERATION: "FULLY FARASHED"



After we got rid of our sweeper design; we thought of using a simpler, more robust design, which would collect gamepieces by a clamping motion.

This way it was much more practical to collect gamepieces stabily.

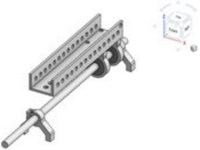


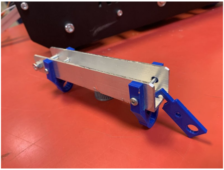
Team #25153 Cartesian
Robotics

PROTOTYPES/ITERATIONS

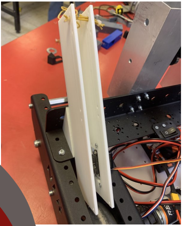
“Plane thrower iterations”

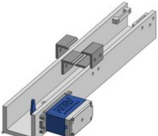
We had multiple designs and ideas about the plane thrower. We started out with a design with a moving part that would hold the plane.

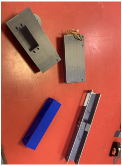




However we ended up in a simpler design that would launch the plane directly by rubber bands. With this design we were able to launch the plane very accurately





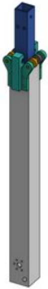


Team #25153 Cartesian Robotics

PROTOTYPES/ITERATIONS

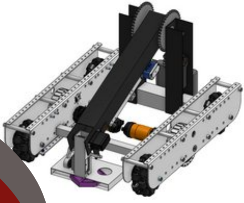
Final iteration

"FULLY FARASHED"

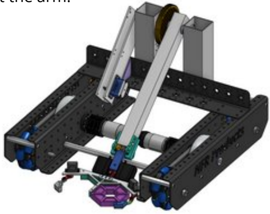


Although we had a great mechanism for collecting gamepieces, we didn't have a great arm mechanism to use it effectively so we designd a telescopic arm.

To retract the telescopic part of the arm, we first though about using a motor but then we wouldn't have enough motors for the lift mechanism so we made a big pulley that had a rope attached to it and attached the pulley to the servo to retract the arm.



G.E.N. V1



"G.E.N."FINAL

PROGRAMMING

PID CONTROLLER OF THE ARM:

When we first thought about this design, we didn't think that the arm would be heavy enough to rotate the motor when it moved. However, when we built the design, the motor couldn't resist the arm and we had to implement PID and feedforward control for our arm.

AUTONOMOUS

In the autonomous section, our robot moves step by step. We wrote functions that would enable us to make our robot drive and turn at values that we can specify. We use these functions in our autonomous period to control our robot's movement and complete our autonomous objectives.

Our goal in the autonomous process is to leave the purple pixel on the line and then the yellow pixel on the backboard. We use TensorFlow object detection to detect our team prop and trained a custom model with the FIRST Machine Learning Toolchain. Our camera detects whether the prop is on the left, middle or right and acts accordingly.

LIFT

We first tried to use a mechanism connected to our intake arm for our lift system, but due to reasons such as the durability of the arm and the sensitivity of the intake, we decided to make a completely separate system connected to the chassis of the robot.

For this purpose, we decided to make a system consisting of hooks and ropes. But a problem arose, the power of our engine was not enough to lift the robot. In order to solve this problem, we designed a gearbox that provides loss of speed but increases torque and printed this model with our 3D printer.

After the Istanbul Off-Season, we had to make changes in the lift due to problems with the robot holding the bar. One of such changes was to change the supports for the lift from acrylic to polycarbonate.

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